Notes on incorporating new AviationDB data into pilot study
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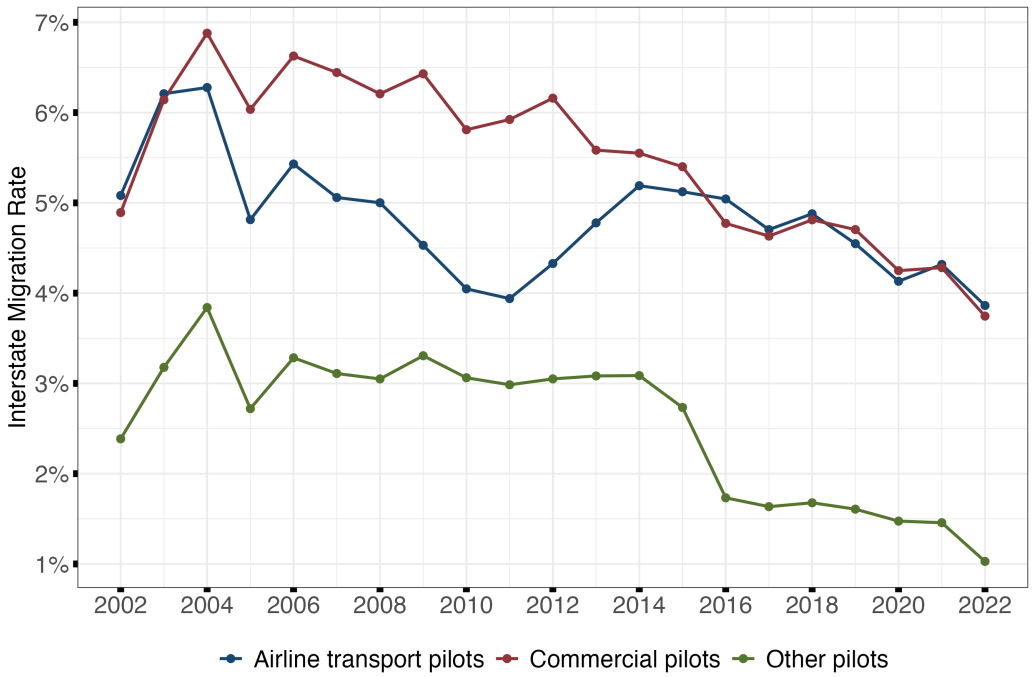
1 Motivational Evidence

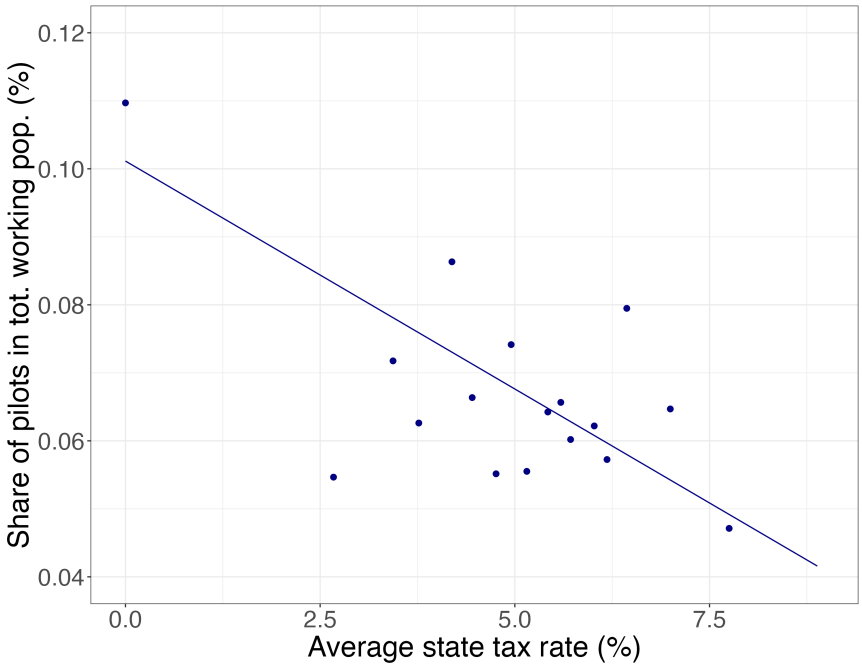
1.1 Pilots are Mobile

- Airline transport pilots and commerical pilots have much higher interstate migration rates (4-7%) compared to the general population (2-3%)

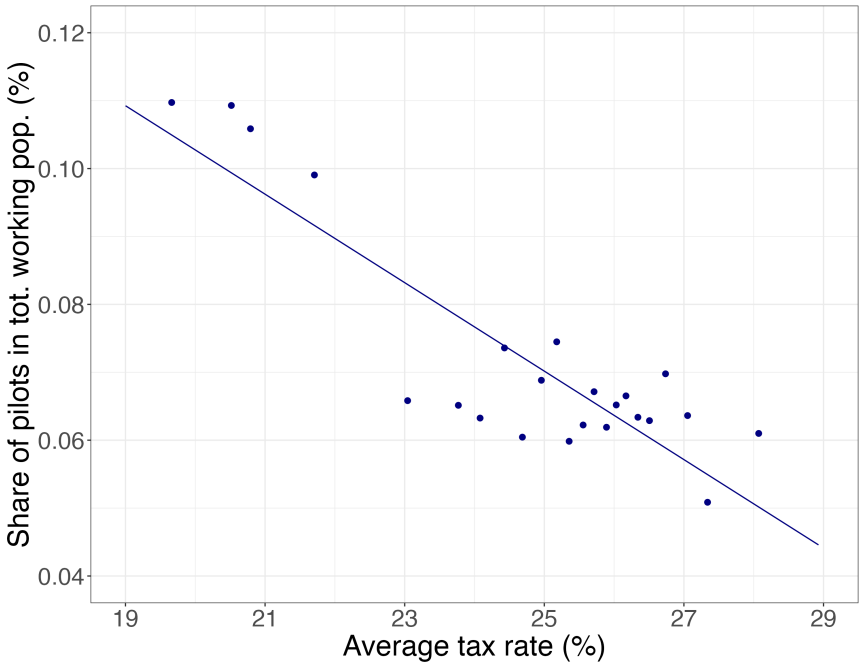


2 Correlating Taxes with Pilot Density

- Binscatter plots showing that lower tax states have a higher concentration of pilots, mainly driven by zero-income tax states
- x-axis: average state tax rate (top), average state + federal tax rate (bottom)
- y-axis: Share of pilots with respect to total working population in each state by year



Notes: tax measure simulated using mean wage of airline pilots, 2007-2022 sample



Notes: tax measure simulated using mean wage of airline pilots, 2007-2022 sample

3 Empirical Strategy

3.1 Stock Model (Agrawal et. al 2025)

Goal: Do states with lower taxes have a larger population of pilots?

$$\ln\left(\frac{N_{s,t}}{N_{s',t}}\right) = \beta \ln\left(\frac{1-\tau_{s,t}}{1-\tau_{s',t}}\right) + \gamma_s + \gamma_t + u_{s,t}$$

where

- s indexes state, s' indexes the reference state, and t indexes year.
- LHS: population of pilots in state s relative to the reference state
- RHS: net-of-income-tax rate in state s relative to the reference state
- γ_s and γ_t: state and year fixed effects

Table 1: AFM stock model results: Balanced panel

	log_stock_n				
	BLS mean	BLS median	SOI p90	SOI p95	SOI p99
	(1)	(2)	(3)	(4)	(5)
lnet_atr	2.718**	2.674**	-1.256	-1.131	0.7594
	(1.099)	(1.081)	(1.865)	(2.022)	(1.026)
Observations	784	784	1,078	1,078	1,078
R^2	0.99759	0.99758	0.99614	0.99613	0.99613
Within R^2	0.02698	0.02626	0.00628	0.00428	0.00396
year fixed effects	✓	✓	✓	✓	✓
dest_state fixed effects	✓	✓	✓	✓	✓

Notes:

- The effects go away if I use the AGI thresholds from IRS SOI individual tax stats to construct the tax measures.

3.2 Migration Log Odds Ratio (Moretti and Wilson 2017)

Goal: Do destination-origin state tax differentials affect the probability of moving from the destination state to the origin state?

Regression:

$$\underbrace{\log\left(\frac{P_{odt}}{P_{oot}}\right)}_{\substack{\text{migration}\\ \text{log odds ratio}}} = \beta_0 + \beta_1 \underbrace{[\log(1-\tau_{dt}) - \log(1-\tau_{ot})]}_{\substack{\text{log dest-origin}\\ \text{net-of-tax diff.}}} + \underbrace{\gamma_t}_{\substack{\text{year}\\ \text{FE}}} + \underbrace{\gamma_d}_{\substack{\text{destination}\\ \text{state FE}}} + \underbrace{\gamma_o}_{\substack{\text{origin}\\ \text{state FE}}} + \underbrace{\gamma_{od}}_{\substack{\text{state-pair}\\ \text{FE}}} + u_{odt}$$

where o indexes the origin state, d indexes the destination state, and t indexes year.

- LHS: migration log odds ratio, between the number of pilots who moved from the origin state o to the destination state d and the number of pilots who stayed in the origin state o in year t.
- RHS: log net-of-average-personal-income tax rate difference between the destination state and the origin state.

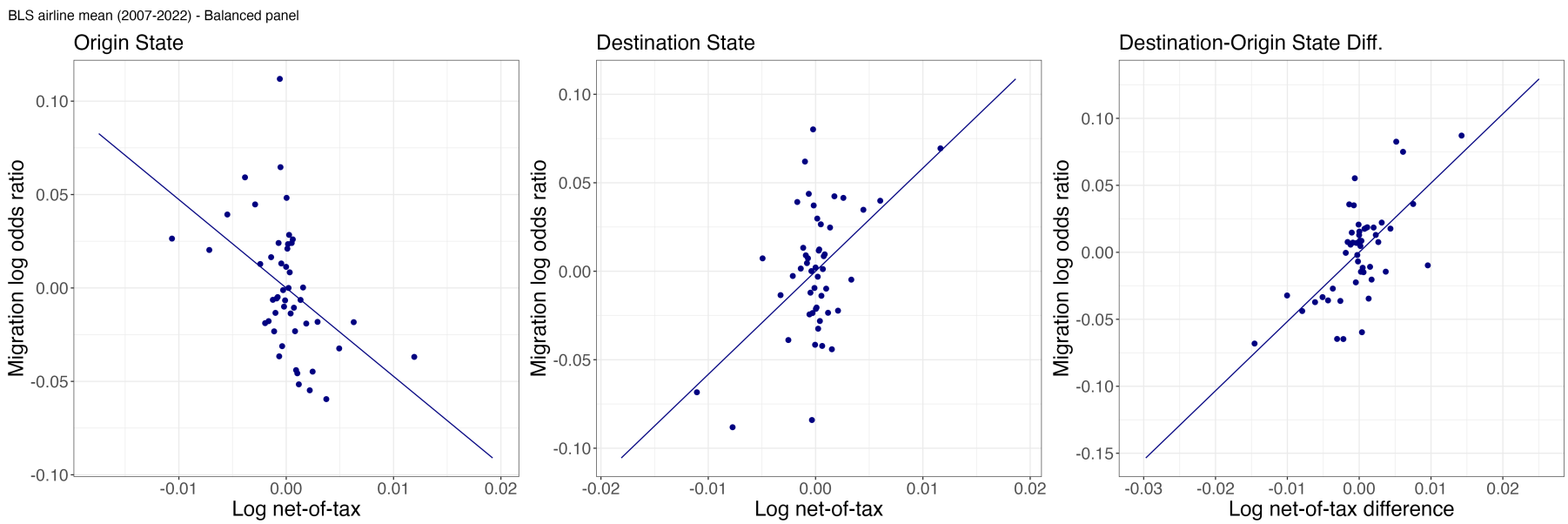


Figure 1

Table 2: Moretti and Wilson (2017) Regression Specification Results, 3-way clustering: Balanced panel

	lodds_ratio	
	BLS airline mean (2007-2022)	BLS airline median (2007-2022)
	(1)	(2)
lnet_tax_diff	5.168***	5.287***
	(1.292)	(1.309)
Observations	9,109	9,109
R^2	0.83683	0.83685
Within R^2	0.00322	0.00333
year fixed effects	✓	✓
origin_state fixed effects	✓	✓
dest_state fixed effects	✓	✓
origin_state-dest_state fixed effects	✓	✓

3.3 Individual-Level Linear Probability Model

Goal: Are pilots less likely to move out of the origin state if the net-of-tax rate is high?

Regression:

$$\text{Moved}_{i,o,t} = \beta(1 - \tau_{o,t}) + \gamma_o + \gamma_t + \gamma_i + u_{i,o,t}$$

- LHS: 1 if moved, 0 if not.
- RHS: net-of-ATR in origin state
- γ_o, γ_t, and γ_i: origin state, year, and pilot fixed effects

Dependent Variable:	moved	
Model:	(1)	(2)
Variables		
net_origin_atr	-0.0070***	-0.0101**
	(0.0015)	(0.0044)
Fixed-effects		
year	Yes	Yes
origin_state	Yes	Yes
unique_id		Yes
Fit statistics		
Observations	101,037	101,037
R^2	0.00696	0.13874
Within R^2	8.93×10^{-5}	0.00020

Clustered (origin_state & unique_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

3.4 Quasi-Experimental Design: Event Study

This is still in progress. The idea is to do a DID utilizing the 2017 TCJA, or to do a staggered DID utilizing major state income tax changes (defined using a threshold, say 1pp). If successful, this should be the main empirical strategy.

Current challenges:

- There are only 5 major state income tax changes (above 1pp) during 2007-2022.
- Still need to figure out a reasonable control group for airline pilots. Previously the idea is to use commercial pilots as a control group, but based on the migration rates graph, commercial pilots seems to be similarly mobile, despite earning much less than airline pilots.